Machine Learning
Homework 4

Dec. 2025

Fill your Name
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1. Here is a sample table. Please use the data in the table 1 to solve the decision tree-related problems.

Day	**Outlook**	**Temperature**	**Humidity**	**wind**	**Play Tennis**
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Hot	High	Strong	No

Table 1: Instances for building a decision tree.

(a) Compute the Overall Entropy.

(b) Compute the information gain and select the root node of the decision tree.

Ans:

(a) The training set contains 14 instances, among which: $|Yes| = 9$, $|No| = 5$. Thus, the empirical entropy of the dataset is

$$H(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} = 0.940. \tag{1}$$

.

(b) The information gain of each candidate attribute is summarized as follows:

- Outlook: conditional entropy:
$H(S|Outlook) = \frac{5}{14}(-\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.693$,
information gain:
$Gain(S, Outlook) = 0.940 - 0.693 = 0.247$.
- Temperature: conditional entropy:
$H(S|Temperature) = \frac{4}{14}(1.00) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$,
information gain:
$Gain(S, Temperature) = 0.940 - 0.911 = 0.029$.
- Humidity: conditional entropy:
$H(S|Humidity) = \frac{7}{14}(0.985) + \frac{7}{14}(0.592) = 0.789$,
information gain:
$Gain(S, Humidity) = 0.940 - 0.789 = 0.151$.
- Wind: conditional entropy:
$H(S|Wind) = \frac{8}{14}(0.811) + \frac{6}{14}(1.00) = 0.892$,
information gain:
$Gain(S, Wind) = 0.940 - 0.892 = 0.048$.

Since **Outlook** achieves the largest information gain, it is selected as the root node of the decision tree.

2. Here are the results of three classifier ensembles, where C denotes the base classifier and C^* denotes the ensemble result. The three subplots correspond to three typical error distribution patterns of base classifiers.

Table 2: Ensemble 1

	t_1	t_2	t_3
C_1	✓	✓	×
C_2	×	✓	✓
C_3	✓	×	✓
C^*	✓	✓	✓

Table 3: Ensemble 2

	t_1	t_2	t_3
C_1	✓	✓	×
C_2	✓	✓	×
C_3	✓	✓	×
C^*	✓	✓	×

Table 4: Ensemble 3

	t_1	t_2	t_3
C_1	✓	×	×
C_2	×	✓	×
C_3	×	×	✓
C^*	×	×	×

(a) Please indicate which ensemble methods (Boosting / Random Forest / Bagging) can improve these error scenarios.

Ans:

(a) As shown in the figure, three base classifiers C_1, C_2, C_3 are evaluated on three test samples t_1, t_2, t_3. A check mark (✓) indicates a correct prediction, while a cross (×) indicates an incorrect one. The classifier C^* represents the final ensemble prediction obtained by a voting-based combination rule. The three subfigures illustrate different patterns of prediction errors, which can be used to explain the core principles of Bagging, Boosting, and Random Forest.

The left subfigure corresponds to a typical successful scenario of Bagging (Bootstrap Aggregating). It can be observed that different base classifiers make errors on different samples, while for each test sample, the majority of classifiers produce correct predictions. As a result, majority voting leads to a perfectly correct ensemble prediction. This phenomenon reflects strong error complementarity among base learners. By training models on different bootstrap samples of the training data, Bagging reduces the correlation among high-variance learners, such as decision trees, thereby effectively reducing variance and improving generalization performance.

The middle subfigure illustrates a degenerate or failure case that may occur in Boosting. In this case, all base classifiers make identical predictions on all samples, resulting in completely overlapping errors. Consequently, the ensemble classifier C^* performs no better than any single base classifier. This situation highlights an important limitation of Boosting: if base learners lack diversity or if subsequent learners fail to correct the mistakes of previous ones, Boosting cannot continue to reduce bias and may degenerate into a single-model predictor. This example emphasizes that Boosting relies on weak learners with correctable error patterns to be effective.

The right subfigure helps explain the motivation behind Random Forest. Here, each base classifier performs well on a subset of samples but poorly on others, and no single classifier is uniformly strong. However, due to the diversity in their error patterns, ensemble voting can still yield improved performance. Random Forest extends Bagging by introducing additional randomness in the feature space: at each split, only a randomly selected subset of features is considered. This structural randomness further reduces correlation among trees and systematically enhances diversity, leading to more stable and robust ensemble predictions.

In summary, the three subfigures collectively reveal the fundamental principle of ensemble learning: performance gains arise not from highly accurate individual models, but from diversity among base learners. Bagging primarily reduces variance through data resampling, Boosting aims to reduce bias via adaptive reweighting of samples, and Random Forest achieves a strong balance by combining data and feature randomness to construct effective ensembles.

3. Perform clustering analysis on the following two-dimensional data using the k-medoids algorithm. Assuming k=2, initialize the cluster centers using data points G and H. Calculate the similarity between any two points using Manhattan distance.

ID	A	B	C	D	E	F	G	H
feature 1	1.2	2.1	1.9	2.3	3.1	4.2	1.2	3.3
feature 2	1.1	2.4	3.1	4.2	4	4	1	2.2

Table 5: Instances for the k-medoids

(a) Provide the clustering results and key computational steps.

Ans:

(a) First, calculate the distance from each data point to the cluster center:

- D(AG)=|1.2-1.2|+|1.1-1|=0.1, D(BG)=2.3, D(CG)=2.8, D(DG)=4.3, D(EG)=4.9, D(FG)=6.
- D(AH)=3.2, D(BH)=1.4, D(CH)=2.3, D(DH)=3, D(EH)=2, D(FH)=2.7.

Therefore,

- **Cluster 1**: A, G
- **Cluster 2**: B, C, D, E, F, H

Computer new cluster centers

- Sum(A, G) = 0.1
- Sum(B-C, D, E, F, H)=10.6, Sum(C-B, D, E, F, H)=10, Sum(D-B, C, E, F, H)=9.6, Sum(E-B, C, D, F, H)=8.8, Sum(F-B, C, D, E, H)=12.8, Sum(H-B, C, D, E, F)=11.4

So, the update of cluster centers are: G, E. Following this approach, the second update cluster is:

- **Cluster 1**: A, B, G (cluster center: A)
- **Cluster 2**: C, D, E, F, H (cluster center: B)

Subsequent cluster centers remain unchanged, and the update is complete.